

SIMATS ENGINEERING

CO1- ASSESSMENT TOOL 3

Problem-Solving Assessment

COURSE NAME: ANALOG AND DIGITAL COMMUNICATION COURSE CODE: ECA08

COURSE OUTCOME COVERED

CO 1: Understand and analyse the principles of communication systems including block diagrams, frequency division multiplexing, amplitude modulation (AM) with DSB-SC, SSB-SC, and VSB-SC generation and detection, the role of VSB in analog television transmission, and AM receiver architectures with key parameters

(Bloom Level mapped-BL4)

Topics Covered

Basic Block Diagram of a Communication System-Frequency Division Multiplexing (FDM) for Broadcasting-Amplitude Modulation (AM)-Principles, Double Sideband (DSB-SC) and Single Sideband (SSB-SC)-Generation and Detection -Vestigial Sideband Modulation (VSB-SC)-Principle and Generation-VSB's critical role in Analog Television Video Transmission, AM Receiver Architectures-Receiver Parameters

Assessment Tool number : CO1-AT3

Assessment Tool Weightage : 40%

Assessment Tool Name : Problem-Solving Assessment

Duration of this assessment : -

Marks : 100

Type of Assessment conduction: Self Learning

- The assessment is a self-learning assessment and it's done as a home task.
- The faculty will provide a set of 20 analytical and numerical problems related to the course topics.
- Students will solve the problems individually within the stipulated time.
- Problems may involve derivations, calculations, performance analysis, and application of communication engineering concepts.
- Students are expected to apply appropriate formulas, engineering principles, and standard procedures to obtain solutions.

- Step-by-step calculations, assumptions, and justifications shall be clearly presented.
 - The assessment aims to enhance analytical thinking, quantitative problem-solving ability, and application of engineering principles to communication system problems.
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Dr.

QUESTION BANK:

Total Marks: 5 marks each

- 1) An AM transmitter radiates 25 kW when the carrier is unmodulated and 29 kW when sinusoidally modulated. Determine the modulation index and percentage modulation. If a second modulating signal producing 40% modulation is added simultaneously, calculate: Total modulation index, Total radiated power, Total sideband power. Mention a practical communication application where this transmission scheme is used.
- 2) A sinusoidal carrier voltage of frequency 2 MHz and amplitude 120 V is amplitude modulated by a sinusoidal voltage of frequency 8 kHz producing 60% modulation. Determine: Lower sideband frequency, Upper sideband frequency, Amplitude of each sideband, Draw the spectrum. Also explain a practical application of AM sideband generation.
- 3) An audio signal $20 \sin \sin (2\pi \times 1000t)$ modulates a carrier $80 \sin(2\pi \times 2 \times 10^5 t)$. Calculate: Modulation index, Sideband frequencies, Amplitude of each sideband, Bandwidth, Transmission efficiency. Mention a suitable communication application.
- 4) An AM DSBFC transmitter operates with carrier power; $P_c = 200W$ and modulation coefficients: $m_1 = 0.2, m_2 = 0.5, m_3 = 0.4$. Determine: Total modulation coefficient, USB power, LSB power, Total transmitted power. Mention a suitable communication application.
- 5) A carrier frequency of 25 MHz is amplitude modulated by a 5 kHz signal. If $m_a = 0.7$ and carrier amplitude is 10 V, determine: Upper sideband frequency, Lower sideband frequency, Amplitude of each sideband. Sketch the spectrum.
- 6) The output voltage of a transmitter is $v(t) = 400(1 + 0.5 \sin \sin 6280t) \cos \cos (6.28 \times 10^7 t)$ applied to a 500Ω load. Determine: Carrier frequency, Message frequency, Carrier power, Total transmitted power. State a practical application.
- 7) An AM wave is represented by $8(1 + 0.7 \cos \cos 3140t) \sin \sin (2\pi \times 10^6 t)$. Determine: Maximum amplitude, Minimum amplitude, Frequency components, Amplitude of each component. Mention a practical application.
- 8) The output voltage of a transmitter is given by $v(t) = 300(1 + 0.5 \sin(9420t)) \cos(9.42 \times 10^7 t)$. This voltage is fed to a 500Ω resistive load. Determine: Carrier frequency, Message frequency, Carrier power, Mean power output (total transmitted power). Also mention a practical communication application where such an amplitude-modulated signal is employed, emphasizing its importance for wide-area coverage and compatibility with simple envelope detector receivers.

9) a) Consider sinusoidal modulation in an AM system. Assuming no over-modulation, the maximum and minimum values of the envelope are 12 V and 4 V respectively. Determine the modulation index.

b) A 250 W carrier is simultaneously modulated by two audio signals with modulation percentages of 50% and 30% respectively. Calculate: Effective modulation index, Total transmitted power. Also explain the significance of this AM transmission in broadcasting systems.

10) An amplitude modulator has the output (in volts) $S(t) = A\cos(800\pi t) + B\cos(760\pi t) + B\cos(840\pi t)$. The carrier power normalized to a 1 Ω resistance is 72 W. The ratio of the total sideband power to the total transmitted power is $\frac{P_{SB}}{P_T} = \frac{1}{10}$. Determine the value of B(in volts), rounded off to two decimal places.

11) The output voltage of a transmitter is given by $v(t) = 100(1 + 0.4\sin(628t))\sin(1.256 \times 10^6 t)$. This voltage is fed to a 500 Ω resistive load. Determine: Carrier frequency, Message frequency, Carrier power, Mean power output (total transmitted power).

12) a. An AM broadcast station transmits a total power of 12 kW when the carrier is modulated by a sinusoidal signal with a modulation index of 0.8. Determine: Carrier power, Transmission efficiency

b. The load current in the transmitting antenna of an unmodulated AM transmitter is 10 A. Calculate the antenna current when the modulation index is 60%.

13) For an AM Superheterodyne Receiver with an intermediate frequency (IF) of 455 kHz, a radio frequency (RF) of 1000 kHz, and a loaded quality factor $Q = 100$, determine the local oscillator frequency, image frequency, and image frequency rejection ratio (IFRR).

14) A Superheterodyne receiver operates in the frequency range of 88 MHz to 108 MHz. The intermediate frequency f_{IF} and local oscillator frequency f_{LO} are chosen such that $f_{IF} < f_{LO}$. It is required that the image frequencies lie outside the 88 MHz–108 MHz band. Calculate the minimum required intermediate frequency f_{IF} (in MHz).

15) For a Superheterodyne receiver, the intermediate frequency (IF) is 10.7 MHz and the local oscillator frequency is 120 MHz. If the frequency of the received signal is greater than the local oscillator frequency, determine the image frequency (in MHz).

16) For a Superheterodyne receiver, the intermediate frequency is 25 MHz and the local oscillator frequency is 2.8 GHz. If the received signal frequency is greater than the local oscillator frequency, determine the image frequency (in MHz).

17) An AM radio station broadcasts at a carrier frequency of 1500 kHz with a maximum modulating signal frequency of 8 kHz. The transmitted power is 20 kW with a modulation index of 0.6. The receiver is located 75 km away in a suburban area with low interference.

Calculate:

- The bandwidth required for the AM signal.
- The carrier power and total sideband power.
- The transmission efficiency of the AM system.

18) A DSB-SC (Double Sideband Suppressed Carrier) system transmits voice signals with a maximum frequency of 6 kHz. The carrier frequency is 300 kHz, and the transmitted power is 10 W.

Calculate:

- The bandwidth required for the DSB-SC signal.
- The total transmitted power in both sidebands.
- The Signal-to-Noise Ratio (SNR) at the receiver if the received power is 5 mW and the noise power is 0.1 mW.

19) A digital TV broadcasting system uses Vestigial Sideband (VSB) modulation with a video signal bandwidth of 6 MHz and a carrier frequency of 55 MHz. The transmitted power is 15 kW, and the receiver is located 75 km from the transmitting station. Calculate the total bandwidth required for the VSB signal, assuming a vestigial sideband width of 0.75 MHz. Determine the bandwidth efficiency of VSB compared to conventional AM transmission, and analyze the received power at the receiver location assuming free-space propagation conditions.

20) A superheterodyne receiver is designed for FM broadcasting with an input frequency range of 76 MHz to 90 MHz. The intermediate frequency (IF) is set at 10.7 MHz, and the receiver employs a single-conversion architecture. Calculate the local oscillator frequency required to receive a station operating at 82 MHz. Determine the corresponding image frequency.

21) The output voltage of a transmitter is given by $v(t) = 400(1 + 0.4\sin(5000t))\sin(2\pi \times 6 \times 10^7 t)$. This voltage is applied to a 600Ω resistive load. Determine: Carrier frequency, Message frequency, Carrier power, Mean power output (total transmitted power).

22) The output voltage of a transmitter is given by $v(t) = 800(1 + 0.6\sin(3000t))\sin(2\pi \times 1 \times 10^8 t)$. This voltage is applied to a 750Ω resistive load. Determine: Carrier frequency, Message frequency, Carrier power, Mean power output (total transmitted power).

23) a) Consider a sinusoidal amplitude modulation (AM) system. Assuming no over-modulation, the maximum and minimum values of the envelope are 15 V and 7 V, respectively. Determine the modulation index.

b) A carrier of 180 W is simultaneously modulated by two audio signals with modulation percentages of 40% and 80%, respectively. Calculate the effective modulation index and the total radiated power.

24) A DSB-SC transmitter is used to transmit audio signals with a maximum frequency of 8 kHz. The carrier frequency is 40 kHz, and the total transmitted power is 80 W. Determine: The bandwidth required for the DSB-SC signal. The power contained in the upper sideband (USB). The power contained in the lower sideband (LSB). The total transmitted power.

25) A DSB-SC transmitter is employed to send speech signals with a maximum frequency of 6 kHz. The carrier frequency is 50 kHz, and the total transmitted power is 100 W. Determine: The bandwidth required for transmission. The power in the upper sideband. The power in the lower sideband. The total transmitted power.

26) A carrier of 20 MHz is amplitude modulated by a sinusoidal signal of frequency 5 kHz and amplitude 10 V. If the modulation index is $m_a = 0.5$, Determine: Sketch the frequency

spectrum of the AM waveform. Calculate the upper sideband (USB) and lower sideband (LSB) frequencies. Determine the amplitude of each sideband.

27) A carrier of 10 MHz is amplitude modulated by a sinusoidal signal of frequency 3 kHz and amplitude 12 V. If the modulation index is $m_a = 0.8$, Determine: Sketch the frequency spectrum of the AM waveform. Calculate the upper and lower sideband frequencies. Determine the amplitude of each sideband component.

28) A carrier of 25 MHz is amplitude modulated by a sinusoidal signal of frequency 6 kHz and amplitude 15 V. If the modulation index is $m_a = 0.4$, Determine: Sketch the frequency spectrum of the AM waveform. Calculate the frequencies of the upper and lower sidebands. Determine the amplitude of each sideband.

29) A DSB-SC system transmits voice signals with a maximum frequency of 6 kHz. The carrier frequency is 30 kHz, and the transmitted power is 80 W. Calculate the bandwidth required for the DSB-SC signal. Determine the total transmitted power in both sidebands. Analyze the SNR at the receiver if the received power is 25 mW and the noise power is 0.5 mW.

30) A DSB-SC communication system transmits audio signals with a maximum frequency of 5 kHz. The carrier frequency is 50 kHz, and the transmitted power is 100 W. Calculate the required bandwidth for the DSB-SC signal. Determine the total transmitted power contained in the upper and lower sidebands. Evaluate the SNR at the receiver when the received signal power is 40 mW and the noise power is 0.8 mW.

31) A DSB-SC transmitter is used to send speech signals with a maximum frequency of 8 kHz. The carrier frequency is 60 kHz, and the transmitted power is 120 W. Determine the bandwidth required for transmission. Calculate the total power transmitted in both sidebands. Analyze the SNR at the receiver if the received power is 30 mW and the noise power is 0.6 mW.

32) An audio frequency signal $80 \sin \sin (2\pi \times 1000t)$ is used to amplitude modulate a carrier signal $160 \sin \sin (2\pi \times 2 \times 10^5t)$. Calculate: Modulation index, Upper and lower sideband frequencies, Amplitude of each sideband component, Bandwidth of the AM signal, Transmission efficiency.

33) In an AM communication system, a message signal of amplitude 50 V and frequency 3 kHz modulates a carrier signal having amplitude 250 V and frequency 500 kHz. Determine: The modulation index, The frequencies of the upper and lower sidebands, The amplitude of each sideband, The transmission bandwidth, The transmission efficiency.

34) A medium-wave broadcasting station uses an audio signal represented by $60 \sin \sin (2\pi \times 4000t)$ to modulate a carrier signal represented by $300 \sin (2\pi \times 10^6t)$. Determine: The modulation index of the AM wave, The frequencies of the upper and lower sidebands, The amplitude of each sideband component, The bandwidth required for transmission, The transmission efficiency of the AM system.

35) An AM DSBFC transmitter operates with an unmodulated carrier power $P_c = 200 \text{ W}$ and is simultaneously modulated by three sinusoidal signals having modulation coefficients $m_1 = 0.30, m_2 = 0.40, m_3 = 0.50$. Determine: The total coefficient of modulation. The individual powers of the Upper Sideband (USB) and Lower Sideband (LSB). The total transmitted power.

36) A medium-wave AM broadcasting station transmits an unmodulated carrier power of 250 W. The carrier is simultaneously modulated by three audio signals with modulation indices of 0.20, 0.45, and 0.60 respectively. Calculate: The effective modulation coefficient. The power contained in the USB and LSB. The total radiated power.

37) An AM DSBFC transmitter has a carrier power of $P_c = 180\text{ W}$ and is simultaneously modulated by three sinusoidal signals whose modulation coefficients are $m_1 = 0.35, m_2 = 0.25, m_3 = 0.55$. Determine: (i) The total coefficient of modulation. (ii) The individual powers of the Upper Sideband (USB) and Lower Sideband (LSB). (iii) The total transmitted power.

38) An AM transmitter radiates 20 kW of power when the carrier is unmodulated and 24 kW of power when the carrier is sinusoidally modulated. Determine: The modulation index. Percentage modulation. If a second sinusoidal modulating signal corresponding to 30% modulation is added simultaneously, calculate: Total modulation index, Total radiated power, Total sideband power.

39) An AM broadcasting station transmits 25 kW of carrier power when unmodulated. When modulated by a single sinusoidal signal, the total transmitted power becomes 30.625 kW. Determine: The modulation index. Percentage modulation. If an additional modulating signal producing 50% modulation is transmitted simultaneously, calculate: Effective modulation index, Total transmitted power, Total sideband power.

40) An AM transmitter radiates 12 kW when the carrier is unmodulated and 14.16 kW when modulated by a sinusoidal signal. Determine: The modulation index. Percentage modulation. If another sinusoidal signal producing 60% modulation is added, calculate: Total coefficient of modulation, Total radiated power, Total sideband power.

Code of Conduct:

I _GODDAVULA NITISH KANTH (192512232) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: NITISH KANTH.G

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation